

N.1

$$\sum_{i=1}^n i^3 = \frac{n^2 (n+1)^2}{4} \quad \forall n \geq 1$$

P.B. $1 = \frac{1}{4}$ ok

P.I.

$$\sum_{i=1}^{n+1} i^3 = \frac{n^2 (n+1)^2}{4}$$

$$\frac{n^2 (n+1)^2}{4} + (n+1)^3 = \frac{(n+1)^2 (n+2)^2}{4}$$

$$\frac{n^2 (n+1)^2 + 4(n+1)^3}{4} = \frac{(n+1)^2 (n+2)^2}{4}$$

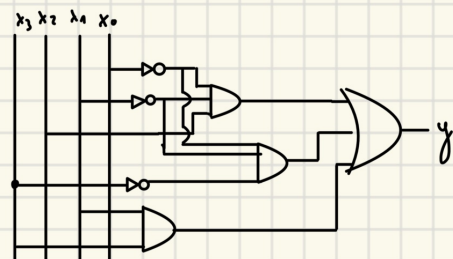
$$\frac{(n+1)^2 [n^2 + 4n + 4]}{4} = \frac{(n+1)^2 (n+2)^2}{4}$$

N.2

x_0	x_1	x_2	x_3	y
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

$x_0 x_1$ \ $x_2 x_3$	00	01	11	10
00	1			
01		1	1	
11	1	1	1	
10	1			

$$x_1 x_3 + \overline{x_0 x_1 x_2} + \overline{x_0 x_1 x_3}$$



N.3

$$\forall x [P(x) \vee Q(x)] \rightarrow \exists x [P(x) \wedge Q(x)]$$

D: $\mathbb{N}$

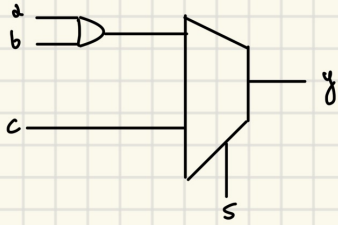
$P(x)$ = x è pari

$Q(x)$ = x è dispari

$$\neg (\neg \exists x \neg [P(x) \vee Q(x)]) \vee \exists x [P(x) \wedge Q(x)]$$

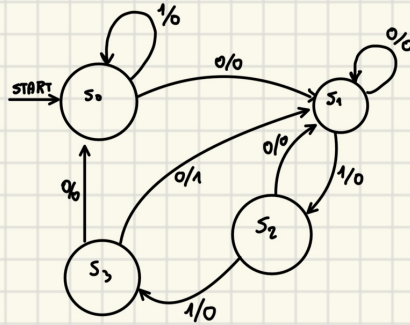
$$\neg (\forall x [P(x) \vee Q(x)]) \vee \neg \forall x \neg [P(x) \wedge Q(x)]$$

N. 4 $y = ab + a\bar{b}c + \bar{a}c$



N. 5

Q_0	Q_1	x	D_0	D_1	y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	0	1	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	1	0	0	0



Q_0Q_1	00	01	11	10
x				
0	1	1	1	1
1				1

$$D_0 = \bar{Q}_0 Q_1 x + Q_0 \bar{Q}_1 x$$

$$D_1 = x + Q_0 \bar{Q}_1$$

